HANNAH KIM, PH.D., M.S.

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last updated: 2026-11-13

RESEARCH INTERESTS & PROFILE

- Postdoctoral research associate studying evolutionary genomics of cancer, with a focus on reconstructing cancer phylogeny at single-cell resolution.
- Methods developer for comparative sequence analysis: property-informed evolutionary models (PRIME) in the HyPhy/Datamonkey ecosystem.
- Interdisciplinary background spanning bioinformatics and human-computer interaction, with a research program in bioinformatics education and tutorials in the era of generative AI.
- Community builder experienced at initiating and facilitating collaborations across disciplines and career stages.

EDUCATION

PhD in Bioinformatics, Temple University — Aug 2019 - Aug 2026
Advisor, Bioinformatics: Dr. Sergei L. Kosakovsky Pond
Co-Advisor, Human-Computer Interaction (HCI): Dr. Stephen MacNeil
Dissertation Topic: Harnessing interdisciplinarity in the investigation of nucleotide sequence evolution
[Recap | Abstract | Shorts | Dissertation]

MS in Computational Biology, Carnegie Mellon University — Aug 2015 - Dec 2016
Relevant Coursework: Computational Genomics, Machine Learning, and Algorithms & Advanced Data Structures

BS in Chemistry, Carnegie Mellon University — Aug 2010 - May 2013
Relevant Coursework: Principles of Computing, and Modern Analytical Instrumentation

TECHNICAL SKILLS

AI-Assisted Research Workflows: LLM-based coding agents for pipeline prototyping and literature triage, prompt-driven pipeline prototyping
Languages: Python, R, Bash, JavaScript, Go
Bioinformatics & Statistics: HyPhy / Datamonkey (comparative sequence analysis, phylogenetics), single-cell RNA-seq analysis, differential gene expression, gene ontology & network analysis, machine learning for genomics
Tools & Infrastructure: Git/GitHub, Google Cloud Platform, reproducible workflow automation, LaTeX
Education & Outreach Tools: Galaxy Training Network, retrieval-augmented generative-AI e-tutorial development, IRB-approved user-study design

SELECTED PUBLICATION

Kim, H., Scheffler, K., Nekrutenko, A., Martin, D. P., Weaver, S., Murrell, B., Kosakovsky Pond, S. L. (2026). The Physicochemical Basis of Protein Evolution: Property-Informed Evolutionary Models (PRIME). *Molecular Biology and Evolution*. doi:10.1093/molbev/msag200 [Preprint | Code | Web App | Data]

Huzar, J., **Kim, H.**, Kumar, S., Miura, S. (2022). MOCA for integrated analysis of gene expression and genetic variation in single cells. *Frontiers in Genetics*, 13:831040. doi:10.3389/fgene.2022.831040

Ding, Y., **Kim, H.**, Madden, K., Loftus, J., Chen, G., Allen, D., Zhang, R., Xu, J., Chen, C., Xu, Y., Tasian, S., Tan, K. (2021). Network Analysis Reveals Synergistic Genetic Dependencies for Rational Combination Therapy in Philadelphia Chromosome-like Acute Lymphoblastic Leukemia. *Clinical Cancer Research*, 27(18). doi:10.1158/1078-0432.CCR-21-0553

Kim, H., Kosakovsky Pond, S. L., MacNeil, S. (2025). Conversations over Clicks: Impact of Chatbots on Information Search in Interdisciplinary Learning. *2025 IEEE Frontiers in Education Conference (FIE).* doi:10.1109/FIE63693.2025.11328556
- FIE 2025 [Preprint | Teasers | Slide | Video | Recap]

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate Sep 2026 - present
University of Mississippi – Sayaka Miura lab *Remote*

- Reconstructing cancer phylogeny at single-cell resolution by integrating evolutionary models with single-cell multi-omics data.

Project Lead Feb 2024 - May 2026
Temple University ACME Group & Temple University HCI lab *Philadelphia, PA*

- Identified gaps in knowledge and proposed ideas that can make data and method work together.
- Designed a user study, drafted a protocol approved by the institutional review board, and ran the study whose results we reported in *Conversations over Clicks* (Kim et al., 2025).

Bioinformatics Engineer Oct 2021 - Jan 2022
Lifetime Omics *Remote*

- Automated cutting-edge methods and analyzed COVID-19 metagenomics data in the Google Cloud environment.
- Developed detailed standard operating procedure for reproducibility.

Bioinformatics Analyst/Software Engineer Jul 2017 - Jun 2019
Children's Hospital of Philadelphia – Kai Tan lab *Philadelphia, PA*

- Investigated RNA-Seq and microarray data from B-Cell Acute Lymphoblastic Leukemia subtypes using differential gene expression analysis, Gene Ontology enrichment analysis, and other relevant bioinformatics methods.
- Identified cancer-specific genetic interactions that led to publications in high-impact journals.

TEACHING EXPERIENCE

Co-Coordinator/Reflective Practicum Completion for THE Certificate May 2023 - Jul 2023
Temple University *Deep Learning for the Life Sciences Journal Club*

- Co-organized a 10-week summer journal club with Dr. Enzo Carnevale for researchers of diverse levels of expertise (undergraduate students, graduate students, postdocs, alumni, visitors, and professors); wrote up the lessons learned on maintaining engagement across disparate prior knowledge in *The Journal for Research and Practice in College Teaching* (2023).
- Group size: 35.

Teaching Assistant Aug 2020 - Dec 2020
Temple University *Genomics in Medicine (BIOL-3111/5111)*

- Generated formative and summative assessment materials and provided timely feedback (Youtube/Introduction).
- Class size: 150.

Teaching Assistant Aug 2019 - May 2020
Temple University *Wet Lab Courses (BIOL-2112 and BIOL-1012)*

- Gave a short lecture in the beginning of every lab, monitored student performance, and provided guidance.
- Graded lab reports and generated quizzes.
- Class size: 20 (× 2 sections) each.
  - BIOL-1012 General Biology II, Spring 2020 was an introductory wet lab course for non-biology majors.
  - BIOL-2112 Introduction to Cellular and Molecular Biology, Fall 2019 was a lab for biology majors.